

Evaluating Reinforcement Learning Agents in Hexxed Game Using Neuro-Nav Toolkit

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1 What is Reinforcement Learning (RL)?

In artificial intelligence and computer science, two contrasting paradigms characterize how computers demonstrate learning: Connectionism and Good Old-Fashioned Artificial Intelligence (GOFAI). The currently dominant paradigm, the ‘bottom-up’ approach of connectionism, involves analyzing information from its elemental components and gradually building a model, employing techniques like gradient descent and fine-tuning model parameters for optimal results. On the other hand, GOFAI, often called ‘symbolic AI,’ relies on explicit, predefined rules and representations to perform tasks, emphasizing symbolic manipulation and reasoning using predefined knowledge structures. By extending the problem-solving abilities of these approaches, reinforcement learning aims not just to solve a problem but also to do so in a way that maximizes a particular reward.

Human learning, however, is a bit more enigmatic. Humans engage in a process more analogous to top-down processing, starting with a broader context and guiding the interpretation of details based on prior knowledge and expectations [1]. Despite a remarkable body of work studying cognitive abilities, there is still a gap in our understanding of how humans seamlessly transition between mental models as they develop a skill. This transition remains a subject of ongoing research and exploration within cognitive neuroscience.

Algorithms such as gradient descent have proven to be highly effective in leveraging the computational power available in modern machines and optimizing models with a finite set of parameters. However, they face challenges when dealing with complex problems that involve a vast and possibly infinite set of parameters. In such scenarios, search algorithms emerge as powerful problem-solving tools [2]. Search algorithms are designed to explore large solution spaces and efficiently find optimal or near-optimal solutions. They are particularly well-suited for tasks that require navigating many possibilities, such as combinatorial optimization or game-playing strategies. By intelligently exploring the solution space and efficiently narrowing down possibilities, search algorithms offer an invaluable approach to tackling intricate problems beyond traditional optimization methods’ capabilities. As technology advances and computational power grows, integrating search algorithms with existing optimization techniques holds tremendous promise for addressing increasingly complex challenges across various domains.

A behavior like this, while intuitive to living organisms, is yet to be mathematically described with the efficiency of biological agents. For example, this behavior is analogous to animals foraging for food in the wild. As animals seek resources from berry patches of unknown rewards to survive, learning involves trial and error as individuals explore and refine their approaches to find the most effective ones for various tasks. Search algorithms and foraging theory share principles of exploration and exploitation. Both optimize solutions, whether animals find abundant food or computers efficiently solve complex problems. By incorporating observed foraging principles, search algorithms become more adaptive and efficient in navigating vast solution spaces.

In the field of cognitive neuroscience, the study of human behavior often involves examining and selecting from established tasks. Lab-based tasks, while offering a controlled environment for studying specific cognitive processes, often simplify many real-life complexities (Appendix A). On the other hand, mastery of real-life skill-based tasks, such as sports or musical instruments, requires decades of dedicated practice to achieve expertise. The complexity of these tasks showcases the human brain’s incredible adaptability and learning potential in response to challenges and continuous improvement. In this context, Hexxed, a mobile strategy game designed to collect information, offers a unique opportunity to study human decision-making, bridging the gap between lab-based tasks and real-life complexities.

2 Introduction to Hexxed

Hexxed, a unique mobile strategy game, was developed in-house by Dr. Gautam Agarwal and Tiago Quendera. This game, designed to provide a real-life, small (but sufficiently complex) decision space, offers a novel approach to studying human skill learning. The mobile game medium not only enables us to understand and collect data on a large and diverse population but also provides a platform for studying human decision-making in a way that was not possible with previously established lab-based tasks.

Upon beginning level 1 of the game, a user is presented with six triangles (henceforth referred to as “lobes”) arranged into a hexagon in the center of the screen, with minimal instructions reading “CONTROLS: TAP AND SWIPE” (Figure 1). Above the instructions are three empty bars that users learn to associate with points earned, missed, and net progress toward beating the level. One randomly selected lobe has a gray stripe towards the edge, which players learn to be a hint showing where the target will appear.

The game’s objective is to maximize the target, as indicated by the size of a triangle occupying the target lobe, and then collect the points from the said lobe. Hexxed is built with six levels of gradually increasing difficulty - level one has only one target, and the number of targets is increased linearly per level all the way up to six targets.

Remarkably, despite the exploratory and adaptive behavior exhibited in the early stages, players demonstrate consistent and surprising behaviors as early as level 1, despite the immense learning curve of the unfamiliar interface. As outlined in the subsequent sections, these behaviors persist and manifest across future levels. This suggests that level 1 data provides insight into the heuristics of solving complex problems like the Knapsack Problem (described more below).

2.1 Previous Findings

The task presented in Hexxed can be considered a representation of the NP-hard Knapsack Problem. The Knapsack Problem is a classic optimization problem in computer science and mathematics in which an agent is given a set of items,

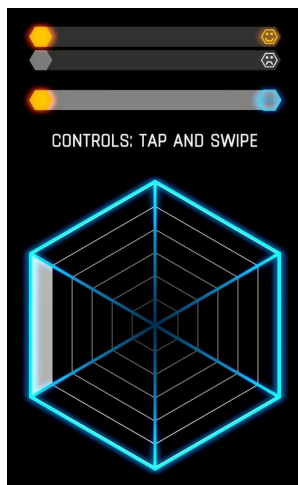


Figure 1: The Hexxed mobile game board

each with a specific weight and value. The agent’s objective is to maximize the total value of items in the knapsack with a fixed weight capacity without exceeding the target weight. In Hexxed, we present users with a finite set of actions (i.e., tapping and swiping) and ask them to sample from combinations of actions in a sequence. In this way, each sequence, carrying a particular earned reward and the weight of points missed, can be considered an item to put in the “knapsack” to beat the level. A defining feature of the class of NP-hard problems is that they can all be efficiently reduced to each other in the same class (More precisely, “efficiently” here means we can reduce one problem to another in at most polynomial time; see Appendix B for the hierarchy of complexity classes). In other words, if we can determine an efficient solution to just one of these problems, we can replicate the algorithm to solve the most challenging problems in computer science. To help find an algorithmic solution, studying human problem-solving patterns may be able to provide some clues. Below, I’ve outlined a few of the most striking discoveries of the way humans solve the task.

2.1.1 People are “picky.”

In the face of countless options and possibilities, we have observed that individuals employ a limited set of preferred strategies: they are “picky.” This limitation could stem from prior successes or learned behaviors from similar previously played games. While this pickiness can be advantageous as it simplifies decision-making, it may also lead to missed opportunities or suboptimal outcomes in scenarios where unexplored strategies could yield better results.